

The axis integral as a convergent coefficient series

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Abstract

The first genuinely infinite coefficient identity of the analytic bridge (Astra #4, rank 4). If the one-variable amplitude has a convergent expansion $\Psi(u, s) = \sum_i c_i(s) u^i$ on $[0, b]$ with the majorant $|c_i(s)| \rho^i \leq H(s)$ for some $\rho > b$, then the axis integral of unit 87 is the absolutely convergent series

$$I_\Psi(s) = \int_0^b \frac{\Psi(u, s) - \Psi(0, s)}{u} du = \sum_{i \geq 1} \frac{b^i}{i} c_i(s),$$

and, when $|w|H$ is integrable against the Gaussian log weight, the constant coefficient of the $d = 2$ equal-exponent block is the absolutely convergent series

$$B = \frac{\log B_*}{k_1 k_2} \mathcal{M}_{p,0}(c_0) - \frac{1}{k_1 k_2} \mathcal{M}_{p,1}(c_0) + \frac{1}{k_2} \sum_{i \geq 1} \frac{b^i}{i} \mathcal{M}_{p,0}(a_i) + \frac{1}{k_1} \sum_{j \geq 1} \frac{b^j}{j} \mathcal{M}_{p,0}(b_j),$$

with a_i, b_j the Taylor coefficients of $\Phi(u, 0, s)$ and $\Phi(0, v, s)$. No convergence of the inverse- N expansion is asserted. Zero sorry/axiom.

1 The things formalised

```
theorem coeff_geometric_bound (c H b : ℝ) (i : ℕ) (hb : 0 < b) (h : 0 < )
  (hmaj : |c| * b ^ i ≤ H) : |c| * b ^ i ≤ H * (b / ) ^ i
theorem hasSum_zero_eq (c : ℝ → ℝ) (Ψ0 : ℝ)
  (h : HasSum (fun i => c i * (0 : ℝ) ^ i) Ψ0) : Ψ0 = c 0
theorem hasSum_shift_div (c : ℝ → ℝ) (Ψu Ψ0 u : ℝ) (hu : u > 0)
  (h : HasSum (fun i => c i * u ^ i) Ψu) (h0 : Ψ0 = c 0) :
  HasSum (fun i => c (i + 1) * u ^ i) (u ^ 1 * (Ψu - Ψ0))
theorem axisIntegral_hasSum (b : ℝ) (hb : 0 < b) (hb : b < ) (Ψ : ℝ → ℝ → ℝ)
  (c : ℝ → ℝ → ℝ) (H : ℝ → ℝ) (s : ℝ)
  (hΨ : u ↦ Icc (0 : ) b, HasSum (fun i => c i s * u ^ i) (Ψ u s))
  (hmaj : i, |c i s| * b ^ i ≤ H s) :
  HasSum (fun i : ℕ => b ^ (i + 1) / ((i : ℝ) + 1) * c (i + 1) s) (axisIntegral b Ψ s)
theorem axis_series_dominated ... :
  (Summable fun i : ℕ => |b ^ (i + 1) / ((i : ℝ) + 1) * c (i + 1) s|)
  ' i : ℕ, |b ^ (i + 1) / ((i : ℝ) + 1) * c (i + 1) s| ≤ (b / ) / (1 - b / ) * H s
theorem axisIntegral_logMoment_hasSum (p b : ℝ) (hb : 0 < b) (hb : b < )
  (Ψ : ℝ → ℝ → ℝ) (c : ℝ → ℝ → ℝ) (H : ℝ → ℝ) (hc : i, Measurable (c i))
  (hΨ : s, 0 < s → u ↦ Icc (0 : ) b, HasSum (fun i => c i s * u ^ i) (Ψ u s))
  (hmaj : s, 0 < s → i, |c i s| * b ^ i ≤ H s)
  (hH : IntegrableOn (fun s => |momentWeight p 0 s| * H s) (Ioi 0)) :
  HasSum (fun i : ℕ => b ^ (i + 1) / ((i : ℝ) + 1) * logMoment p 0 (c (i + 1)))
  (logMoment p 0 (axisIntegral b Ψ))
```

```

theorem axisB_series ... :
  axisB b p k k Ψ
    = 1 / ((k : ) * k) * (Real.log (b ^ (k + k)) * logMoment p 0 (c 0)
      - logMoment p 1 (c 0))
      + 1 / (k : ) * ' i : , b ^ (i + 1) / ((i : ) + 1) * logMoment p 0 (c (i + 1))
theorem twoDB_series ... :
  twoDB b p k k Φ
    = 1 / ((k : ) * k) * (Real.log (b ^ (k + k)) * logMoment p 0 (a 0)
      - logMoment p 1 (a 0))
      + 1 / (k : ) * ' i : , b ^ (i + 1) / ((i : ) + 1) * logMoment p 0 (a (i + 1))
      + 1 / (k : ) * ' j : , b ^ (j + 1) / ((j : ) + 1) * logMoment p 0 (c (j + 1))

```

2 Mathematics

At $u \in (0, b]$ the integrand is the shifted series $(\Psi(u) - \Psi(0))/u = \sum_i c_{i+1} u^i$ (`hasSum_nat_add_iff'` and $\Psi(0) = c_0$ by uniqueness of sums). Each term integrates to $c_{i+1} b^{i+1}/(i+1)$, and the L^1 norms are bounded by $|c_{i+1}| b^{i+1} \leq H(b/\rho)^{i+1}$, a convergent geometric series; Mathlib's `hasSum_integral_of_summable_integral_norm` then exchanges sum and integral. The log-moment statement is the same exchange one level up, through unit 98's `hasSum_logMoment_of_dominated` with the pointwise domination $\sum_i |b^{i+1} c_{i+1}(s)/(i+1)| \leq \frac{b/\rho}{1-b/\rho} H(s)$. The series formulas for B_Ψ and for the $d = 2$ constant coefficient B are then rewrites of their definitions (units 87 and 88), using $\Psi(0, \cdot) = c_0$ under the log moments.

3 Paper-facing meaning

This is the precise form of Astra #3(c)/#4(c): each coefficient of the finite expansion that is a regularised axis integral equals an absolutely convergent series in the corner Taylor coefficients, under a radius-margin hypothesis $b < \rho$ with a uniform majorant. It is the first formal instance of the paper's "coefficients as convergent series" claim, for an already formalised coefficient (B in the $d = 2$ block). Analyticity enters only as a provider of the majorant; the finite expansion itself needs no analyticity. Nothing is claimed about convergence of the expansion in $N^{-1/k}$, which is generally divergent.

4 Lean notes

- In `fun i => b ^ (i + 1) / ((i : ) + 1) * ...` the ascription `(i : )` makes the binder `real`; write `fun i : => ...` (symptom: type mismatch `i + 1 : ,` or `Countable` failing).
- `integrableOn_const` needs its `autoParam measure_Ioc_lt_top.ne` passed explicitly.
- Pointwise goals from `ae_restrict_iff'` and `setIntegral_congr_fun` are beta-redexes: `beta_reduce` before `rw`.
- `Integrable.const_mul` produces an `And`; ascribe `IntegrableOn` before `.congr_fun`.
- `hasSum_single 0` plus `HasSum.unique` identifies the constant term of a power series at 0 without any analytic machinery.